

Luka D. Stojilković

Software Engineer, Belgrade, Serbia

stojilkovic.d.luka@gmail.com +381 64 904 0342 lukastojilkovic.github.io/CV github.com/lukastojilkovic LinkedIn

Final-year Computer Science student building full-stack products, native Windows apps and local LLM systems. Led a 25-person team on a full-stack banking platform and shipped four open-source Windows apps with installers.

Experience

Software Developer, AI in AEC May 2025 – Jan 2026
AI consulting startup for architecture, engineering and construction

- Built **Neostack**, a prototype web application that helps firms catalogue their software stack and licence usage.
- Built **AI in AEC Flow**, an internal platform for time, task and revenue tracking.

Team Lead, BANKA_2 2025 – 2026
Software Engineering capstone, RAF

- Led a 25-person team delivering an enterprise banking platform: Spring Boot services, a React 19 web client, a Kotlin / Jetpack Compose Android app and Docker infrastructure.
- Wrote the majority of the codebase, covered by 2,165 backend tests (JUnit, Mockito), 1,254 frontend tests (Vitest) and a Cypress end-to-end suite.
- Designed **Arbitro**, a self-hosted LLM banking assistant with tool calling, multi-step wizards, RAG over the specifications and SSE streaming on Ollama.

Selected Projects

- **Nexus**: offline-first desktop workspace (tasks, calendar, notes, finance) in an Electron and React monorepo with an encrypted SQLite store and optional end-to-end encrypted sync. Open-source release in preparation. (TypeScript, Electron, SQLCipher, Supabase)
- **Stem Agent**: LLM agent that specializes itself by evolving pipelines of typed tools with six mutation operators, beam search and a MAP-Elites archive; successful pipelines persist across sessions. (Python, LM Studio, Gemma)
- **PermaDel 1.0**: secure file shredder for Windows with File Explorer integration and Windows Hello verification. Overwrites alternate data streams and flushes every pass; forensic tests scan a raw disk image. (C#, WinUI 3, C++)
- **Refreshify 1.1**: runs 16 built-in Windows repair and cleanup tools in a safe order after creating a restore point, and recognizes and fixes 15 common failures. (C#, WinUI 3)
- **PerfWindow 0.10**: real-time hardware monitor: a Rust dashboard fed by an on-demand .NET sensor service that exits with the window, plus an installer that handles drivers and elevation. (Rust, egui, .NET 8)
- **Serbian real-time speech recognition**: fine-tuned Wav2Vec2 XLS-R with a hybrid voice-activity detector; about 0.29 WER on conversational speech. Group project. (PyTorch, Hugging Face)

Also: neural networks, Word2Vec and classical ML from scratch in NumPy, a C++17 compiler and stack VM, an NFC business-card Android app and image processing in MATLAB. Full list on the portfolio.

Skills

Languages	Java, Kotlin, Python, TypeScript, C#, C++, Rust, SQL
Backend	Spring Boot, Spring Security, JPA/Hibernate, REST, SSE, FastAPI
Frontend and mobile	React 19, Vite, Tailwind CSS, Zod, Jetpack Compose, Hilt, Room
Testing	JUnit 5, Mockito, Vitest, Cypress, xUnit
AI and ML	PyTorch, TensorFlow, scikit-learn, Hugging Face, Ollama, LM Studio, RAG, tool calling, ChromaDB
Data and DevOps	PostgreSQL, MySQL, Oracle, SQLite, Supabase, Docker, Kubernetes manifests, GitHub Actions
Desktop	WinUI 3, Windows App SDK, Electron, egui, Inno Setup

Education and Languages

BSc Computer Science 2023 – expected 2026
School of Computing (RAF), Union University, Belgrade

Computer Science studies 2018 – 2023
Faculty of Mathematics, University of Belgrade

Languages: Serbian (native), English (full professional proficiency).
Other: Karate black belt (1st Dan), elementary music school (piano), driving licence (B).